HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed, trained, and evaluated text classifiers for fraud and abuse detection using **Python** and **scikit-learn**, achieving an average **0.88 AUC** on labelled abuse corpora across phishing and smishing detection.
- Served ML models inline on the message send path, meeting strict **<90ms** latency budgets by optimizing inference code and deploying via **FastAPI** within a containerized environment.
- Developed Python scripts for data curation and labeling, preparing datasets of approximately **1.5M messages** for model training and offline evaluation against diverse abuse archetypes.
- Implemented a shadow-mode evaluation framework to compare model performance against human-reviewed data for phishing and brand impersonation, reporting precision and recall metrics before enforcement.
- Collaborated with fraud operations teams to integrate detection model outputs into human-in-the-loop review tooling, transforming feedback into actionable training signals.
- Participated in code reviews for the fraud detection stack, contributing to raising overall code quality and adherence to engineering standards for the messaging platform.
- Containerized ML inference services using **Docker** and deployed them to a managed Kubernetes cluster, configuring basic resource limits for predictable performance and rollout.

SKILLS

ML Engineering – Text Classification, Model Training, Model Evaluation, Inference, Drift Monitoring, AUC, Precision, Recall, Calibration, Adversarial ML, LLMs, Supervised Learning, Unsupervised Learning

Backend Development – Python, FastAPI, Flask, Async Programming, REST APIs, Microservices, Java, Spring Boot, Hibernate, JPA

Data Engineering & Analytics – SQL, PostgreSQL, Pandas, NumPy, Data Curation, Data Labeling, ETL, Data Warehousing, Scikit-learn, R

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes, Jenkins, Git, CI/CD, Linux, Shell Scripting

Observability & Operations – Splunk, Dynatrace, P1/P2 Incident Response, SLA Management, Change Management, Production Support

PROJECTS

Messaging Fraud Detection System

- Developed a **Python**-based text classification pipeline to detect phishing and smishing attempts within high-volume message traffic, processing ~5,000 messages per hour.
- Trained and evaluated a **scikit-learn** model on a custom-labelled dataset of **200k messages**, achieving **0.91 AUC** and <100ms inference latency for inline decisioning.
- Implemented adversarial resilience techniques to handle obfuscated text and URL evasion patterns, ensuring detection robustness against evolving abuse tactics.

Customer Onboarding Automation Framework

- Built a **Python** CLI tool to automate the initial setup and configuration for new enterprise customers integrating with a backend platform, reducing manual setup time by **40%**.
- Integrated the tool with customer-provided schemas and configuration files, providing immediate feedback on potential integration issues before full deployment.
- Documented the framework and delivered technical walkthroughs to customer success teams, enabling them to onboard pilot clients more efficiently.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)